

--	--	--	--	--	--	--	--

B.Tech. Degree VIII Semester Regular/Supplementary Examination April 2024

ME 19-205-0810 OPERATIONS RESEARCH (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Solve problems using optimization methods.
 CO2: Develop mathematical skills to approach real life industrial problems.
 CO3: Inculcate ideas to solve problems on Transportation and Queuing models.
 CO4: Import and analyse case studies to solve future problems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) How does simplex method give indication that the solution is inbound?	3	L2	1	1
(b) Discuss the scope and limitations of operations Research.	3	L1	3	2
(c) Establish the difference between the feasible solution and basic feasible solution in simplex method.	3	L3	1	2
(d) Describe the concept of duality in linear programming.	3	L1	3	1
(e) Give a brief account of applications of assignment problems.	3	L3	2	2
(f) Explain stepping stone method in the context of Transportation problems.	3	L3	2	2
(g) What can we have $\lambda = \mu$ in an M/M/1 queuing model?	3	L1	4	3
(h) Explain dynamic programming.	3	L2	4	2

PART B

(4 × 12 = 48)

II.	A company produces 2 types of leather belts, say A and B, A is a superior quality and fetch a profit of ₹12/- belt and B is an ordinary one gives a profit of ₹9/- per belt. Belt A require twice as much time as required by belt B' if all belts were B, the company could produce 500 belts per day, but the supply of leather is sufficient only for 400 belts per day. Only 200 fancy buckles required for type A belt and 350 ordinary buckles for Type B are available per day. Formulate the L.P model and find the combination of number of belts produced for maximum total profit.	12	L3	1	2
-----	---	----	----	---	---

OR

III. (a)	Discuss the areas of application of OR in business and industries	4	L2	1	1
(b)	Solve using Graphical method Maximize $Z = 20 \times 1 + 10 \times 2$	8	L5	1	3

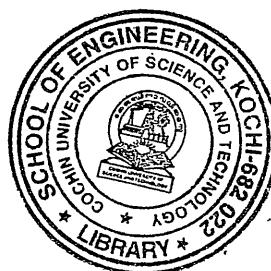
Subject to

$$x_1 + 2x_2 \leq 28$$

$$3x_1 + x_2 \leq 24$$

$$x_1 \geq 2$$

$$x_2 \geq 0$$



(P.T.O.)

- | | Marks | BL | CO | PO | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|-------|----|----|----|---------------|----|----|----|----|-------|----|----|----|----|----|----|----|----|---|----|---|---|----|----|----|----|----|----|------|---|---|---|---|--|
| IV. (a) What is degeneracy in linear programming? | 2 | L1 | 2 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Solve using simplex method
Max $Z = 6x + 4y$
Subject to
$2x + y \leq 390$
$3x + 3y \leq 810$
$y \leq 200$
$x, y \geq 0$. | 10 | L5 | 2 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| OR | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| V. An animal feed company must produce 200 kg of a mixture consisting of ingredients X1 and X2. The ingredient X1 costs ₹3 per kg and X2 costs ₹5 per kg. No more than 80 kg of X1 can be used and at least 60 kg of X2 must be used. Find the minimum cost mixture. | 12 | L5 | 2 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| VI. Solve the following transportation problem | 12 | L4 | 3 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| <table border="1"> <thead> <tr> <th>Orgin/Dest</th> <th>D1</th> <th>D2</th> <th>D3</th> <th>D4</th> <th>Avai.</th> </tr> </thead> <tbody> <tr> <td>O1</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>6</td> </tr> <tr> <td>O2</td> <td>4</td> <td>3</td> <td>2</td> <td>0</td> <td>8</td> </tr> <tr> <td>O3</td> <td>0</td> <td>2</td> <td>2</td> <td>1</td> <td>10</td> </tr> <tr> <td>Req.</td> <td>4</td> <td>6</td> <td>8</td> <td>6</td> <td></td> </tr> </tbody> </table> | | | | | Orgin/Dest | D1 | D2 | D3 | D4 | Avai. | O1 | 1 | 2 | 3 | 4 | 6 | O2 | 4 | 3 | 2 | 0 | 8 | O3 | 0 | 2 | 2 | 1 | 10 | Req. | 4 | 6 | 8 | 6 | |
| Orgin/Dest | D1 | D2 | D3 | D4 | Avai. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| O1 | 1 | 2 | 3 | 4 | 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| O2 | 4 | 3 | 2 | 0 | 8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| O3 | 0 | 2 | 2 | 1 | 10 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Req. | 4 | 6 | 8 | 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| OR | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| VII. (a) What are the different methods used for solving an assignment problem? | 2 | L3 | 3 | 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Solve the following assignment problem for assigning sub-assemblies to contractors to minimize the cost | 10 | L3 | 3 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| <table border="1"> <thead> <tr> <th>Sub Ass/contr</th> <th>C1</th> <th>C2</th> <th>C3</th> <th>C4</th> </tr> </thead> <tbody> <tr> <td>S1</td> <td>8</td> <td>10</td> <td>12</td> <td>16</td> </tr> <tr> <td>S2</td> <td>11</td> <td>11</td> <td>15</td> <td>8</td> </tr> <tr> <td>S3</td> <td>9</td> <td>6</td> <td>5</td> <td>14</td> </tr> <tr> <td>S4</td> <td>15</td> <td>14</td> <td>9</td> <td>7</td> </tr> </tbody> </table> | | | | | Sub Ass/contr | C1 | C2 | C3 | C4 | S1 | 8 | 10 | 12 | 16 | S2 | 11 | 11 | 15 | 8 | S3 | 9 | 6 | 5 | 14 | S4 | 15 | 14 | 9 | 7 | | | | | |
| Sub Ass/contr | C1 | C2 | C3 | C4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S1 | 8 | 10 | 12 | 16 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S2 | 11 | 11 | 15 | 8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S3 | 9 | 6 | 5 | 14 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S4 | 15 | 14 | 9 | 7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| VIII. (a) Write note on arrival pattern and service discipline in the case of queuing theory. | 4 | L2 | 4 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Customer arrive at a clinic at the rate of 8/hour (Poisson Arrival) and the doctor can serve at the rate of 9/hour (exponential) | 8 | L5 | 4 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| (i) What is the probability that a customer does not join the queue and walks into the doctor's room?
(ii) What is the probability that there is no queue?
(iii) What is the probability that there are 10 customers in the system?
(iv) What is the expected number in the system? | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| OR | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| IX. (a) Compare the LPP and Dynamic programming. | 2 | L2 | 4 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) A company has ₹6 lakh to invest. Total three investments are possible. If X_j rupees in lakhs are invested, then the value of V_j is obtained. The expression for $V_j(X_j)$ is as follows: | 10 | L5 | 4 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| (i) $V_1(X_1) = 9X_1 + 4$
(ii) $V_2(X_2) = 5X_2 + 9$
(iii) $V_3(X_3) = 2X_3 + 3$
(iv) $X_1, X_2, X_3 > 0$.
The amount of investment in terms multiples of ₹1 lakh are permitted. How should the company must plan the investment to maximize the profit? | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |